How AI sees

AI · Level 1 · Lesson 3



Today we will

- Explain that a picture is a grid of numbers called pixels.
- Explain how vision AI learns patterns in those numbers from labelled photos.
- Give reasons vision AI can be fooled: light, angle and things it never saw.

Remember

- Every picture is a grid of pixels, and every pixel is a number (three for colour).
- Vision AI learns to pick out edges, then shapes, then whole things, from thousands of labelled photos.
- Light, angles and unfamiliar things change the numbers and can fool it, so check when it matters.

Picture in numbers

1. Draw (5 min): Shade a simple picture on a 10 × 10 grid.
2. Encode (7 min): Write it as numbers: 0 for white, 9 for black.
3. Swap (6 min): A partner shades a new grid from your numbers only.
4. Compare (4 min): Put the two pictures side by side.
5. Fool it (5 min): Change the numbers to “dim the light” and talk about what a vision AI would see.

Each person needs two or three 10 × 10 grids and a pencil. Keep numbers between 0 and 9: when “dimming the light”, stop at 9.

Let’s discuss

1. What would a vision app need to learn before it could recognise a fruit from your region?
2. When has your phone’s camera or face unlock got something wrong?
3. Why should a person check a vision AI’s answer when it really matters?

Exit ticket

- Write a 3 × 3 picture as numbers, and give one reason vision AI can be fooled.